Asset-Light Delivery Network for Nigeria

Dedicated Software and Backend Architecture Design

Document purpose
This document defines the end-to-end software, backend, operational-control, data, API, security, and MVP architecture for an asset-light Nigerian parcel delivery network. It is written without reference to any foreign courier model and is adapted specifically for Nigerian market, infrastructure, regulatory, and trust conditions.

Version	Date	Prepared for	Scope
1.0	May 2026	Asset-light delivery startup planning	Backend, software architecture, workflows, fraud control and MVP roadmap

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1. Executive Summary

The proposed platform is an asset-light parcel delivery operating system for Nigeria. It allows individuals, social-commerce sellers, SMEs and larger business senders to move goods through a controlled network of partner shops, local hubs, city sorting hubs, partner-owned vans and scheduled intercity routes.

The startup does not need to buy vans at launch. It owns the software layer, rules engine, custody ledger, tracking system, hub network rules, pricing logic, fraud controls and settlement process. Partner vans and partner shops provide physical capacity, while the platform provides coordination, trust and visibility.

Core architecture principle
The business is asset-light, but the software must be control-heavy. The platform can avoid owning vehicles, but it cannot avoid owning the operational controls around scanning, custody, routing, fraud detection, pricing, partner settlement and customer tracking.

- Individuals and businesses can send parcels through the same platform.
- Pricing is based on weight, volumetric size, route, delivery type, insurance and bulk volume.
- Hubs serve as drop-off, pickup, failed-delivery and local custody points.
- Partner vans run scheduled intra-city sweeps and intercity trunk routes.
- Bulk movement is allowed, but every parcel must keep its own waybill and traceability.
- Failed doorstep deliveries return to the nearest local hub for receiver collection.
- Every scan is validated against state, city, local area, hub, route and manifest rules.
- Settlement is released only after verified scans, route completion and proof events.

2. Architecture Objectives

Objective	Architecture response
Low capital launch	Use partner shops as hubs and partner-owned vans instead of buying vehicles or building owned depots at launch.
Support personal and business senders	Create sender types, business profiles, bulk upload, address book, wallet and merchant reporting.
Prevent lost goods	Use parcel-level waybills, custody ledger, two-sided handover, seal numbers, weight checks and exception workflows.
Prevent wrong-location shipment	Validate every scan against destination state, city, local area, hub, route and manifest.
Handle weak network conditions	Use offline-first hub and driver apps with local event queues and later sync.
Support Nigerian delivery patterns	Use hubs, scheduled sweeps, intercity trunks, SMS/WhatsApp notifications and optional assisted booking.
Avoid cash-flow traps	Use prepaid sender wallet, limited credit, settlement holds and proof-triggered partner payout.
Build trust	Use KYC, OTP pickup, photo evidence, tracking, claims handling and admin audit logs.

3. Nigeria Context and Design Assumptions

The architecture assumes a market where informal waybill culture already exists, but customers and businesses need stronger tracking, reliability, dispute resolution and proof of custody. The system is not designed as a bike-only on-demand courier app. It is designed as a hybrid pickup/drop-off network with scheduled van movement and controlled doorstep options.

- Internet and power may be unreliable; scanning must work offline and sync later.
- Many receivers may not install an app; SMS and WhatsApp are core notification channels.
- Sellers may use Instagram, WhatsApp, markets, shops or small warehouses.
- Some parcels will be small, but many business users will ship cartons, bulk goods or higher-value items.
- Hubs are convenient but not naturally secure warehouses; controls must compensate for this.
- Partner vans reduce startup capital needs but increase the need for KYC, route discipline and settlement rules.
- Intercity delivery should begin with scheduled corridors, not a nationwide promise from day one.

4. Product Scope and User Segments

User segment	Main needs	System capabilities
Individual sender	Send goods to friends, family, customers or another city.	Simple booking, hub drop-off, payment, tracking and receiver OTP.
Business sender	Send many customer orders and manage returns.	Business profile, bulk upload, wallet, address book, reports, priority support and API later.
Receiver	Know when goods arrive and collect safely.	Tracking page, SMS/WhatsApp alerts, pickup OTP and complaint reporting.
Hub agent	Accept, store, hand over and release parcels.	Hub PWA, scan-in/out, hub inventory, OTP release and exception reporting.
Driver / van partner	Carry goods on assigned routes and get paid reliably.	Route app, manifest scan, proof events, issue reporting and settlement view.
Admin / operations	Control network performance and risk.	Control tower, route management, exceptions, KYC, disputes and settlements.

5. Service Model and Delivery Options

Service	Description	Use case	Pricing position
Hub-to-hub economy	Sender drops at origin hub; receiver collects at destination hub.	Low-cost personal and SME delivery.	Cheapest
Door-to-hub	Platform collects from sender; receiver collects	Busy sellers or businesses that cannot	Medium

	at hub.	leave shop.	
Hub-to-door	Sender drops at hub; final delivery goes to receiver address.	Convenience for receiver.	Medium / premium
Door-to-door	Collected from sender and delivered to receiver address.	Urgent or higher-value deliveries.	Premium
Intercity scheduled	Goods move between city hubs through scheduled trunk routes.	Lagos-Abuja and other corridor traffic.	Route and weight based
Business bulk shipment	Many parcels, cartons or goods move together under a manifest.	SME and marketplace sellers.	Bulk discount with controls

MVP service decision
The first build should support hub-to-hub, hub-to-door, door-to-hub, bulk business shipment and intercity scheduled movement. Full door-to-door should be available only where route density and customer willingness to pay support it.

6. End-to-End Operating Workflows

6.1 Personal Hub-to-Hub Workflow

1. Sender creates shipment through web, app, WhatsApp-assisted form or hub counter.
2. System captures receiver details, destination state, city, local area and preferred hub.
3. System calculates price using weight, size, route and service type.
4. Sender pays or uses wallet balance.
5. System creates waybill number and QR/barcode label.
6. Sender drops parcel at origin hub.
7. Hub agent verifies sender details, checks prohibited-item rules and scans parcel in.
8. Parcel joins a route or manifest after destination validation.
9. Driver collects parcel under two-sided handover.
10. Parcel moves through local/city sorting hub and destination local hub.
11. Receiver gets OTP notification and collects parcel.
12. Hub confirms collection and shipment is completed.

6.2 Business Bulk Workflow

13. Business creates shipments individually, by CSV upload or through merchant API.
14. System validates addresses and groups parcels by state, city, local area and corridor.
15. Each parcel receives its own waybill number.
16. Parcels going to the same destination corridor are added to a bulk manifest.
17. Hub or cross-dock scans parcels into sealed bag, crate or carton.
18. System records manifest ID, seal number, expected parcel count and expected weight.
19. Partner van collects the sealed manifest with two-sided handover.
20. Destination hub checks seal, weight, manifest count and parcel reconciliation.

21. Local hubs receive final parcels for pickup or home delivery.
22. Exceptions are created automatically for missing, wrong-hub, broken-seal or weight-mismatch cases.

6.3 Failed Home Delivery Workflow

23. Driver attempts doorstep delivery.
24. If the receiver is unavailable, address is wrong, phone is unreachable or access is denied, driver records failed delivery reason.
25. Driver uploads GPS, timestamp, note and photo/call evidence where appropriate.
26. Parcel is returned to the nearest approved local hub.
27. Receiver receives SMS/WhatsApp notification with hub address and OTP.
28. Receiver collects from local hub.
29. Local hub completes the delivery record.

7. High-Level System Architecture

```
Sender Web/App/WhatsApp Form     Business Portal     Receiver Tracking Page
        \                |                 /
         \               |                /
          ----------------- API Gateway -----------------
                         |
     -----------------------------------------------------------------
     | Auth | Shipment | Pricing | Hub | Route | Manifest | Scanning |
     | Payment/Wallet | Notification | Fraud/Exception | Settlement |
     | KYC/Compliance | Admin Control Tower | Reporting/Analytics  |
     -----------------------------------------------------------------
        |           |           |           |           |
   PostgreSQL+PostGIS   Redis Queue   Event Log   Object Storage   Maps
        |           |           |           |           |
    Audit trail     Background jobs  Proofs/KYC  SMS/WhatsApp  Payment gateways
```

The API gateway routes authenticated requests from sender apps, business portal, hub PWA, driver app, admin dashboard and receiver tracking page. The backend can start as a modular monolith and later split into services when parcel volume, team size and operational complexity justify it.

8. Core Backend Modules

Module	Purpose	Key outputs
Auth and Identity	Controls users, roles, OTP, sessions and device binding.	Verified user sessions and role permissions.
Shipment Service	Creates and manages parcel records, waybills, statuses and tracking.	Shipment lifecycle and tracking state.
Pricing Service	Quotes delivery price using weight, volume, corridor, service type and add-ons.	Price quote and charge rules.
Hub Management	Manages hub onboarding, service areas, inventory and parcel touch events.	Hub inventory and scan compliance.

Route and Dispatch	Creates urban sweeps, intercity trunks and return-to-hub routes.	Route assignments and trip manifests.
Bulk Manifest	Groups parcels for bulk movement while keeping parcel-level traceability.	Manifest, seal, count and weight records.
Scanning and Custody	Records every scan, handover and custodian change.	Tamper-evident custody ledger.
Payment and Wallet	Handles sender payments, prepaid wallet and refunds.	Payment status and wallet ledger.
Settlement	Calculates hub, driver and van-owner earnings after validation.	Approved settlement records.
Fraud and Exception	Detects misroutes, missing scans, weight mismatch and suspicious activity.	Exception queues and risk flags.
Notification	Sends SMS, WhatsApp, email, push and in-app messages.	Customer and partner notifications.
KYC and Compliance	Stores partner verification, vehicle documents and compliance evidence.	Approved or restricted partner status.

9. Scanning, Custody and Fraud-Control Architecture

The scan system is the operational heart of the platform. A shipment status alone is not enough. The platform must always know who has custody, where the parcel was scanned, what action was taken, and whether the handover was confirmed by both sides.

Control	Rule	Fraud prevented
Two-sided handover	Custody changes only when sender-side and receiver-side custodians confirm.	One person falsely claiming transfer.
Device binding	Hub and driver scans must come from approved devices.	Fake remote scanning.
GPS validation	Hub scans must occur near the registered hub; route scans must match route geography.	Scanning from wrong location.
Photo evidence	High-risk or exception events require photo proof.	False damage, false delivery or false receipt claims.
OTP release	Receivers collect using OTP or approved override.	Wrong receiver collection.
Settlement hold	Partner payout waits for required scan/proof events.	Paying before actual service completion.
Audit log	All admin overrides are logged and reviewable.	Insider manipulation.

10. Destination Validation and Misrouting Prevention

Wrong-location delivery is prevented by treating destination as structured data, not only free text. Every shipment must contain destination state, city, local area, zone, destination hub and fallback local hub.

Validation point	Rule
Adding parcel to manifest	Parcel destination state/city must match manifest destination state/city.
Adding parcel to route	Parcel must be physically at route origin and compatible with route destination.
Receiving at hub	Current hub must match expected destination hub or approved intermediate hub.
Failed home delivery return	Parcel must return to the nearest approved local hub for that receiver zone.
Bulk bag creation	All parcels inside bag must belong to allowed destination group or route corridor.
Admin override	Override requires reason, photo/evidence and audit log.

Example blocked scan
If a parcel is going to Abuja - Wuse Hub but an agent tries to add it to a Lagos-to-Port Harcourt manifest, the backend should block the scan and show: Scan blocked. Destination is Abuja - Wuse Hub. This manifest is for Port Harcourt.

11. Bulk Movement and Manifest Architecture

Bulk scanning helps speed up hub-to-hub and city-to-city movement, but it must not remove individual parcel traceability. A manifest can move as one operational unit, but every parcel inside must remain visible and reconcilable.

Manifest: MAN-LAG-ABJ-0001
Origin: Lagos Sorting Hub
Destination: Abuja Regional Hub
Route: Lagos to Abuja
Vehicle: ABC-123-LA
Driver: Assigned driver
Seal Number: SEAL-99281
Expected Parcel Count: 87
Expected Weight: 320kg
Status: Sealed for Transit
Manifest Items: WB-0001, WB-0002, WB-0003 ... WB-0087

Manifest control	Description
Seal number	Each bag, crate or carton has a unique seal number recorded at origin and checked at destination.
Expected count	System stores expected number of parcels in the manifest.
Expected weight	System stores expected bulk weight and accepted

	variance.
Parcel list	Each parcel in the manifest remains individually tracked.
Reconciliation	Destination hub confirms seal, weight, count and individual parcel presence.
Random recount	System can force full individual scan based on risk level.
High-value rule	High-value goods require individual scans at origin, loading and destination.

12. Failed Home Delivery and Return-to-Hub Flow

Home delivery should not become an uncontrolled retry loop. When delivery fails, the driver returns the parcel to the nearest approved local hub. The local hub then becomes the collection point.

Failed delivery reason	Required evidence	Next action
Receiver unavailable	Call attempt log, timestamp, GPS.	Return to local hub.
Wrong address	Driver note, GPS, optional photo.	Return to local hub and ask receiver/admin to update address.
Phone unreachable	Call attempt log.	Return to local hub.
Receiver refused	Driver note and optional photo/signature.	Return to local hub or start return-to-sender process.
Access denied	Driver note, GPS and photo where appropriate.	Return to local hub.
Weather/security issue	Driver note and route issue record.	Return to local hub or reschedule if approved.

13. Pricing and Wallet Architecture

Pricing must support both single parcels and bulk business shipments. Price is not only a distance fee; it includes handling, route cost, risk, payment cost, insurance provision and margin.

Final Price =
Base Route Fee
+ Hub Handling Fee
+ Actual or Volumetric Weight Fee
+ Doorstep Pickup Fee, if selected
+ Doorstep Delivery Fee, if selected
+ Insurance / Claims Reserve
+ Payment Processing Fee
+ Storage Fee, if parcel overstays free window
- Bulk or Wallet Discount

Pricing input	Purpose
Actual weight	Charges heavy parcels fairly.
Volumetric weight	Charges large but light parcels fairly.

Corridor	Reflects Lagos-Lagos, Lagos-Abuja or other route cost.
Service type	Differentiates hub-to-hub, hub-to-door, door-to-door and intercity.
Declared value	Supports insurance, claims reserve and high-value rules.
Bulk volume	Supports business discounts while retaining traceability.
Storage days	Applies fee after free collection window.
Wallet/prepayment	Rewards prepaid usage and reduces cash-flow risk.

14. Data Model and Database Architecture

The recommended primary database is PostgreSQL with PostGIS for location-aware hub, route and service-area data. Redis should handle job queues, retries and notification dispatch. Object storage should hold KYC files, photos, proof-of-delivery and evidence documents.

Entity group	Tables
Identity and partners	users, businesses, receivers, hub_agents, drivers, van_owners, vehicles, kyc_documents
Network	hubs, hub_service_areas, zones, states, cities, local_areas, routes, route_stops
Shipment	shipments, shipment_events, shipment_items, shipment_addresses
Bulk movement	bulk_manifests, bulk_manifest_items, seal_records, reconciliation_records
Custody and scans	scan_events, custody_events, scan_exceptions, misroute_exceptions
Finance	payments, wallets, wallet_transactions, settlements, refunds
Support and risk	disputes, claims, notifications, audit_logs, prohibited_items

14.1 Key Shipment Fields

shipments
- id
- waybill_number
- sender_user_id
- business_id
- receiver_name
- receiver_phone
- origin_state_id
- origin_city_id
- origin_hub_id
- destination_state_id
- destination_city_id
- destination_local_area_id

- destination_zone_id
- destination_hub_id
- fallback_local_hub_id
- address_text
- weight_kg
- length_cm
- width_cm
- height_cm
- volumetric_weight
- declared_value
- delivery_type
- price
- payment_status
- shipment_status
- current_custodian_type
- current_custodian_id
- created_at
- updated_at

15. API Architecture

API group	Example endpoints
Auth	POST /auth/register, POST /auth/login, POST /auth/verify-otp, POST /auth/refresh-token
Shipments	POST /shipments, GET /shipments/:id, GET /track/:waybillNumber, PATCH /shipments/:id/cancel
Pricing	POST /pricing/quote, GET /pricing/rules, POST /pricing/rules
Hubs	GET /hubs, GET /hubs/:id/inventory, POST /hubs/:id/accept-parcel, POST /hubs/:id/release-parcel
Routes	POST /routes, GET /routes/:id, POST /routes/:id/start, POST /routes/:id/complete
Manifests	POST /manifests, POST /manifests/:id/add-shipment, POST /manifests/:id/seal, POST /manifests/:id/reconcile
Scans	POST /scans/parcel, POST /scans/manifest, POST /scans/validate, GET /scans/exceptions
Payments	POST /payments/initiate, POST /payments/webhook, GET /wallets/me, POST /settlements/:id/approve
Disputes	POST /disputes, GET /disputes/:id, PATCH /disputes/:id/resolve

16. Offline-First Architecture

Hub and driver apps must work during poor data coverage. The app stores scans locally, then syncs when signal returns. The server validates the synced events before accepting them into the authoritative custody ledger.

```
offline_scan_event
- local_event_id
- user_id
- device_id
- shipment_id
- manifest_id
- event_type
- local_timestamp
- gps_latitude
- gps_longitude
- photo_reference
- sync_status
- server_validation_status
```

- Every offline event must include local timestamp, user ID, device ID and GPS where available.
- Server-side validation must check route, hub, destination and custody rules after sync.
- Conflicting events should not silently overwrite each other; they should create exceptions.
- High-risk actions such as receiver release, admin override and high-value handover should require online confirmation where possible.

17. Security, Compliance and Risk Controls

Area	Control
Authentication	JWT access token, refresh token, OTP, role-based access, admin 2FA and device binding.
Data protection	Minimise personal data, encrypt sensitive files, restrict admin access and define retention rules.
Partner KYC	Verify hub owners, drivers, van owners, vehicles, bank accounts and operating documents.
Parcel safety	Prohibited-items list, inspection rights, sender declaration and photo-at-acceptance for selected parcels.
Vehicle compliance	Capture vehicle licence, motor insurance, roadworthiness and driver documents.
Financial control	Prepaid wallet, settlement hold, audit logs and payout only after verified proof events.
Operational risk	Route cut-offs, exception queues, sealed bags, random audits and performance penalties.
Claims	Declared value, liability cap, evidence review and compensation workflow.

18. Admin Control Tower

The admin control tower is the nerve centre of the network. It should not only show dashboards; it should help operations staff act quickly on exceptions.

- Live parcel status by hub, route, city and state.
- Hub inventory and aged parcels.
- Route progress and delayed vans.
- Misroute and wrong-hub scan attempts.
- Broken seal, weight mismatch and missing parcel exceptions.
- Failed home deliveries returned to local hubs.
- Open disputes, claims and compensation requests.
- Hub, driver and van-owner performance.
- Settlement payable and settlement holds.
- Revenue, cost and margin by route.

19. Recommended Technology Stack

Layer	Recommendation
Backend	NestJS with TypeScript or Django. Start as modular monolith; split later only when justified.
Database	PostgreSQL with PostGIS for locations, hubs, routes and service areas.
ORM	Prisma or TypeORM for NestJS; Django ORM for Django.
Queue	Redis with BullMQ for notifications, sync jobs, settlement jobs and retries.
Storage	S3-compatible object storage for KYC, parcel photos, seal photos and proof-of-delivery.
Hub app	Progressive Web App because small shops may prefer browser-based access.
Driver app	React Native or Flutter with offline-first scan queue.
Admin dashboard	Next.js dashboard with control tower, reports and exception queues.
Notifications	SMS, WhatsApp Business API, email and push notifications.
Payments	Paystack, Flutterwave, bank transfer and prepaid wallet.
Maps	OpenStreetMap/OSRM or GraphHopper for routing; commercial map API where required.
Monitoring	Sentry, Prometheus/Grafana or equivalent logging and observability stack.

20. MVP Scope and Pilot Roadmap

20.1 MVP Must-Have Features

- Sender booking for individuals and businesses.
- Weight, size and delivery-type pricing.
- Hub drop-off and hub pickup.
- Hub agent PWA with scanning.
- Waybill number and QR/barcode.
- Parcel tracking page.
- Driver route assignment.
- Parcel scan and manifest scan.
- Destination validation to block wrong hub and wrong manifest scanning.
- Bulk manifest with seal number, expected count and expected weight.
- Payment and wallet support.
- SMS/WhatsApp notification.
- Admin control tower.
- Basic dispute and claims logging.
- Partner settlement report.

20.2 MVP Should Avoid

- Nationwide launch before scan discipline is proven.
- Cold-chain and pharmaceutical delivery until compliance is ready.
- Large owned warehouse system.
- Full crowdshipping as a core promise.
- Cash-on-delivery at scale in the early stage.
- Complex AI route optimisation before operational data is reliable.

20.3 Recommended Pilot

Pilot element	Recommendation
Initial city	Lagos first.
Second city	Abuja.
First corridor	Lagos to Abuja scheduled intercity route.
Hub count	Start with a small controlled hub network, then expand density.
Vehicle model	Partner vans with route-based payout and proof-triggered settlement.
Primary sender segments	SME merchants, Instagram/WhatsApp sellers, offices and individual senders.
Parcel focus	Small and medium parcels, cartons and bulk goods that can be handled safely.
Core KPI target	High scan compliance, low loss/damage, low misroute rate and route-level gross margin.

21. Appendices

21.1 Suggested Shipment Statuses

DRAFT
BOOKED
PAYMENT_PENDING
PAID
DROPPED_AT_ORIGIN_HUB
ACCEPTED_BY_HUB
ASSIGNED_TO_MANIFEST
ASSIGNED_TO_ROUTE
PICKED_UP_BY_DRIVER
ARRIVED_AT_SORTING_HUB
IN_TRANSIT_INTERCITY
ARRIVED_AT_DESTINATION_CITY
READY_FOR_PICKUP
OUT_FOR_DELIVERY
DELIVERY_ATTEMPT_FAILED
RETURNED_TO_LOCAL_HUB
COLLECTED_BY_RECEIVER
DELIVERED
MISROUTED
DAMAGED
LOST
UNDER_INVESTIGATION
CANCELLED

21.2 Suggested Exception Types

WRONG_HUB_SCAN_ATTEMPTED
WRONG_MANIFEST_ATTEMPTED
MISSING_DRIVER_CONFIRMATION
MISSING_HUB_CONFIRMATION
SEAL_BROKEN
WEIGHT_MISMATCH
PARCEL_MISSING_FROM_MANIFEST
HIGH_VALUE_PARCEL_MISSING_SCAN
LOCATION_MISMATCH_SCAN
DEVICE_MISMATCH_SCAN
DUPLICATE_SCAN
FAILED_DELIVERY_EVIDENCE_MISSING
ADMIN_OVERRIDE_REQUIRED

21.3 Implementation Sequence

30. Build core database schema and role-based authentication.
31. Build shipment creation, pricing quote and waybill generation.
32. Build hub PWA scan-in and scan-out.
33. Build driver route view and scan confirmation.
34. Build custody events and shipment events.
35. Build destination validation rules.

36. Build bulk manifest, seal and reconciliation logic.
37. Build payment initiation, wallet and settlement report.
38. Build SMS/WhatsApp notifications.
39. Build admin control tower and exception queue.
40. Pilot with a small number of hubs and partner vans.
41. Expand only after scan compliance and exception handling are stable.